

HealthConnect Clinic – Week 6

Part 3 –Assumptions, Limitations, Risks & Dependencies

1. Assumptions

- The HealthConnect appointment dataset is representative of the appointments being analysed.
- Appointment outcomes such as Attended, No-Show and Cancelled are recorded consistently.
- Booking lead days accurately represent the time between booking and appointment dates.
- The validated KPIs and analytical measures are based on the cleaned Week 5 dataset.
- Relationships identified in the analysis are treated as associations, not proof of causation.
- Small patient/appointment groups are interpreted cautiously.
- The Data Science model findings are used as cross-track supporting evidence and not as a replacement for the Data Analytics analysis.

2. Limitations

Some segments have small numbers of appointments, particularly higher previous-no-show groups and the 60+ booking-lead group.

Small groups can produce unusually high or low percentages that may not generalise.

The analysis is observational, so it cannot establish that booking lead time, reminders or previous no-shows directly cause missed appointments.

Some potentially relevant patient or operational factors may not be available in the dataset.

The Data Science model still has an uncertainty zone, with approximately 36% of test predictions falling between predicted probabilities of 0.4 and 0.6.

Appointment type interacts with other factors such as booking lead time and previous no-show history, meaning single-variable comparisons may not fully explain behaviour.

3. Risks

Risk	Potential Impact	Mitigation
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Small sample sizes in some segments May produce unstable percentages Monitor group sizes and interpret small groups cautiously

Association interpreted as causation Could lead to inappropriate business decisions Clearly distinguish association from causation

Changes to the dashboard or calculations Could introduce inconsistent KPI results Revalidate KPIs after changes

Model uncertainty Could affect expectations around predictive accuracy Continue testing model performance and uncertainty

Over-reliance on one variable Could oversimplify no-show behaviour Analyse interactions between lead time, history, reminders and appointment type

Cross-track integration issues Could cause inconsistent interpretation Document exchanged findings and validate conclusions across tracks

4. Dependencies

The main Week 6 cross-track dependency was:

Data Analytics ↔ Data Science

Data Analytics provided validated findings on:

- booking lead time;
- previous no-show behaviour;
- appointment type;
- reminder status;
- segment relationships;
- validated KPIs.
- Data Science reviewed these findings against its model and error analysis and provided:
- feature-importance information;
- model/error-analysis feedback;
- confirmation of important predictive signals;
- a potential Week 7 modelling direction around appointment type × booking lead time.

5. Issues Resolved During Week 6

Week 5 KPI calculations were revalidated and remained accurate.

Week 5 findings were tested further rather than being accepted without validation.

The Specialist Consultation interpretation was clarified with Data Science.

The Week 5 dashboard was preserved by duplicating it before Week 6 development.

Cross-track integration moved from a planned dependency to an actual analytical exchange.

6. New Issues / Items to Monitor

- Further testing is required for the appointment type × booking lead-time interaction.
- Small groups need continued monitoring.
- The reminder relationship requires further testing before making causal claims.
- Week 7 should test whether the Week 6 findings remain reliable after further validation.

Part 3 Overall Conclusion

The Week 6 review identified no major unresolved data-quality issue that prevents the Data Analytics work from progressing. The main concerns are analytical rather than basic data-quality problems: small segment sizes, potential over-interpretation of associations, model uncertainty and the need to validate interactions further. These risks have been documented and will be carried into Week 7 testing and refinement.